

CRYENGINE®

CRYENGINE 5 Beginner's Guide

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Course Setup

Installing CRYENGINE and GameSDK

This course requires CRYENGINE 5.5 or higher with **GameSDK** from the CRYENGINE marketplace. Here are the steps to install them:

1. Download and install the free CRYENGINE Launcher from here by clicking on the **Download** button. Note that you will need to create a free CRYENGINE account if you don't already have one. If you need additional help with this, refer to our Installation Quick Start guide.
1. Start the Launcher and **sign in** to your CRYENGINE account.
2. In the Launcher, click on **Marketplace** > **Marketplace**. **Search** for "GameSDK" and **add** the asset to your cart. Make sure you add the latest version, not an archived version.
3. **Check out** and **download** the asset from the marketplace.
4. In the Launcher, click on **Library** > **My Assets**. **Click** on the **arrow** next to your **GameSDK** asset and choose **Reveal in Explorer**. You'll be in a folder called **gamesdk_#.#** (where #.# is the software version number) . **Open** the **GameSDK** folder inside this.
5. Back in the Launcher, click on **Library** > **My Projects**. **Click** on the **Import** button. (GameSDK is a complete project, not just an asset, so it needs to be imported as a project.) Browse to the same **gamesdk_#.# > GameSDK** folder you've already opened (not the deepest level GameSDK folder). Click **OK** to **import** your GameSDK project into the Launcher.

Opening the GameSDK Project

Click on the **GameSDK** project from the **Launcher** to open the project.

Once you've launched the CRYENGINE editor, you need to tell it to scan all of your project folders so it becomes aware of the GameSDK assets that you just added. Find the **Asset Browser** tool (if it isn't visible, enable it from the **Tools** > **Asset Browser** menu). In the Asset Browser menu, choose **Edit** > **Generate/Repair All Metadata**. Wait until all of the messages in the top right corner of the screen have disappeared before continuing. The Asset Browser generates a .cryasset file for every asset in your folders. These are required by the asset management system. After this is done, **restart CRYENGINE**.

Course Learning Objectives

Upon completion of this course, students will know how to:

- Navigate the CRYENGINE interface, Editor, and tool set
- Arrange customize tool panels
- Work with Brush entities
- Create and edit materials and textures
- Sculpt, import, and paint terrain, including making roads
- Add and manage vegetation using the Vegetation Editor
- Use Soft Depth Blending to seamlessly blend entities into the terrain
- Use the Environment Editor and Level Settings to control sunlight, time of day, latitude and longitude, and procedural volumetric clouds
- Precisely position and modify objects using snap and alignment tools
- Use assets and functionality from the free GameSDK (Software Development Kit)

Course Workbook Conventions

To help you understand the text in this workbook, we have used the following **formatting** conventions:

Important concepts are highlighted in **bold**, or refer to something you can **do** or **click on**: a button, entity, menu item, etc.

Nested items in **menus** or **tool panels** are **separated** with the **right caret character**, e.g.: **Tools** > **Level Editor** > **Create Object**.

Concepts and goals are explained first, followed by the exact **step by step instructions** to achieve them. Tips about best practices look like this yellow box.

***Pro Tip*:**

This is a "pro tip" suggesting best practices or informing you about something of particular importance.

Introduction

Welcome to the CRYENGINE beginner's guide!

We're going to learn basic CRYENGINE concepts and get familiar with the sandbox editor by setting up a basic level with terrain, vegetation, and brush (non-moving) entities with which the default GameSDK player can inter-

act.

Prerequisites

This course assume zero prior experience with CRYENGINE or game development. You can also view the video tutorial of this course, which comprises just the introduction to our free extensive “flappy boid” beginner’s course, instead of or in conjunction with this course workbook. Our recommendation is that you watch the videos and consult the course book for additional clarification.

You will need to be comfortable navigating the Windows operating system, File Explorer, and using the mouse. Experience with 3D programs is helpful but not required. Dual monitors are also helpful given the number of tools used, but certainly not necessary.

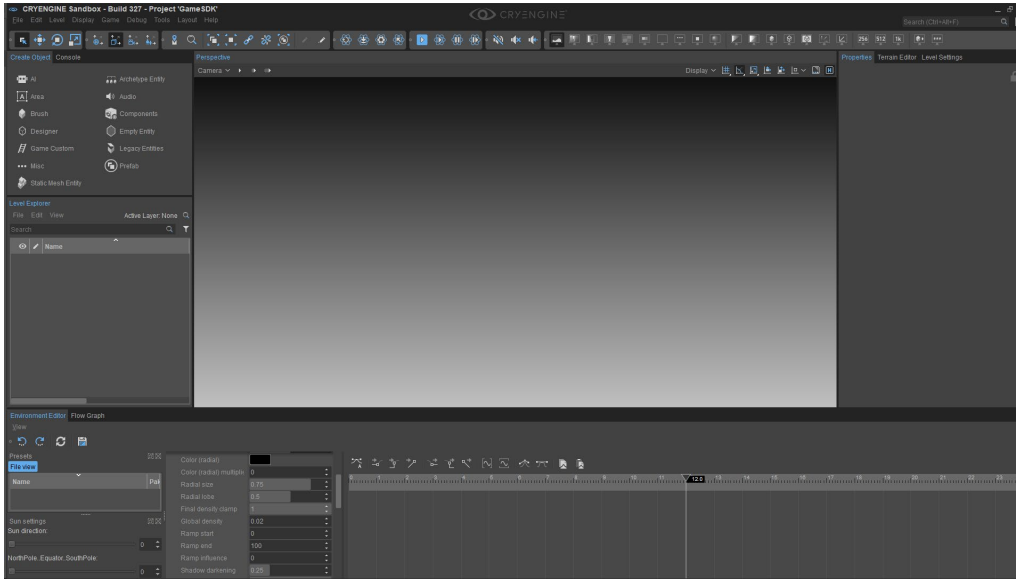
Introduction: Navigation and Tool Layouts

Once you have followed the previous instructions to install CRYENGINE and GameSDK, import the GameSDK project, and open it from the Launcher, the first thing you'll see is the default tool set. What you *won't* see yet is a 3D world, simply because you don't yet have a **level** open. Opening a project is the first step; opening or creating a new level is our next step.

Before we create our first level, let's take a moment to look at how we work with the tool panels. CRYENGINE provides many tools, all of which you'll find in the **Tools** menu. Any tool panel, once visible, can be **moved** by **dragging** its title bar. As you're dragging a tool panel, you'll notice that a series of **boxes** appear to indicate where you can **drop** the tool to **split** an existing tool window horizontally or vertically, or the entire application window. The **middle box** will set the dragged tool as a **tab** in the window where you drop it.

Customizing the Tool Layout

Let's add some tools that we'll be using, and hide others we won't need. **Right-click** on the **Asset Browser** and choose **close**. From the **Tools** menu, choose **Level Editor > Level Settings**. Make it a tabbed window along with Terrain Editor and Properties on the right side. Also make the **Environment Editor** and **Flow Graph** tools visible. Dock them along the bottom of the screen by splitting the entire application window horizontally. Your layout should look something like this screen shot.



If you have more than one monitor, you might want to use it just for a fullscreen Flow Graph/Environment Editor window in this course, as they each present a lot of information. (There are several other tools that are very large as well, so you will find additional monitors to be very useful.)

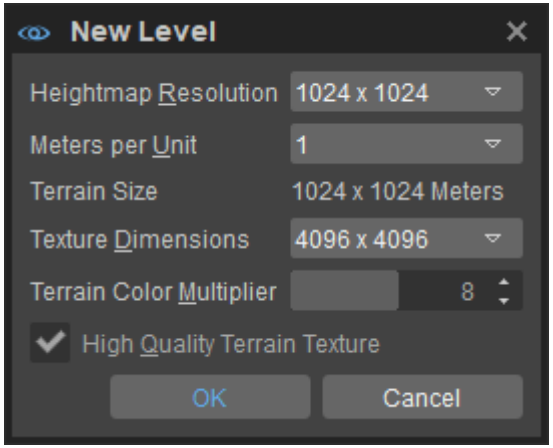
There are also many toolbars. Right-click on any toolbar (just below the main menu) to see them listed. Try enabling the **Physics** and **View Mode** toolbars and hiding the **Layouts** toolbar. Read about customizing toolbar buttons here.

Finally, let's **save** your customized tool layout so you can always go back to it: from the menu, choose **Layout > Save Layout As…** and give it a name.

You'll find a complete description of every item in the main menu here.

Creating a New Level

Create a **new level** called "practice" through **File > New**. (Be sure to keep your levels in the **levels** folder.) Leave the settings at their defaults, as shown here. The heightmap is a grid of points, each with a specific height; this is used to generate the hills and valleys of your terrain, or to keep it flat if you prefer.

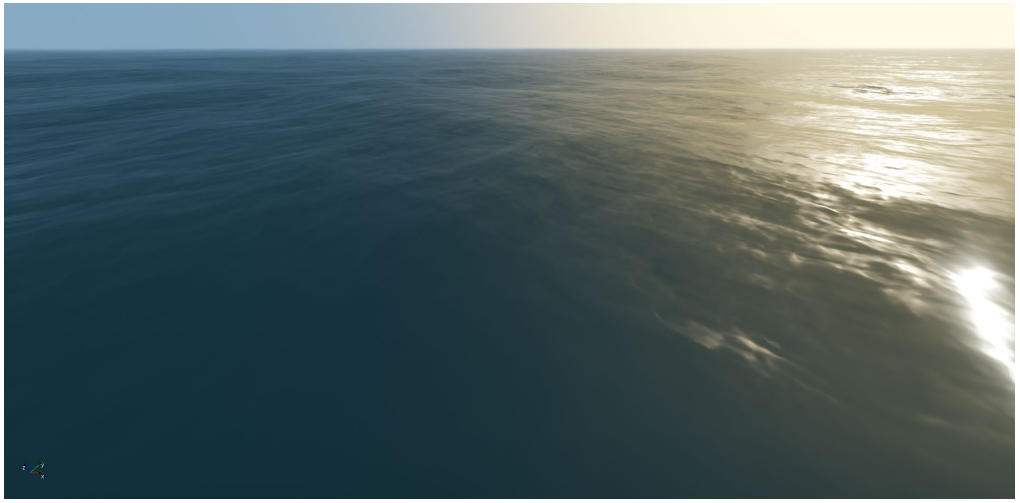


> ***Pro Tip***:
>
> Do not use spaces or non-alphanumeric characters in level names (or anything else); use **underscores** instead of spaces.

After a moment, your first new level will appear in the **Perspective viewport** – a flat grid with the sky and sun above you, and the ocean surrounding you. You are viewing this world through the editor's default camera (as opposed to cameras you'll need to create for your players to use in the actual game).

Navigation in the Editor

By default, you'll see many lines of white technical information displayed in the upper right corner of the Perspective viewport. The first set of numbers at the top are the current **camera position** (Pos) and **rotation** (Angl). The second button from the right toggles through these **Display Info** modes, including a "clean" view with no



information overlay.

Navigation is simple and intuitive:

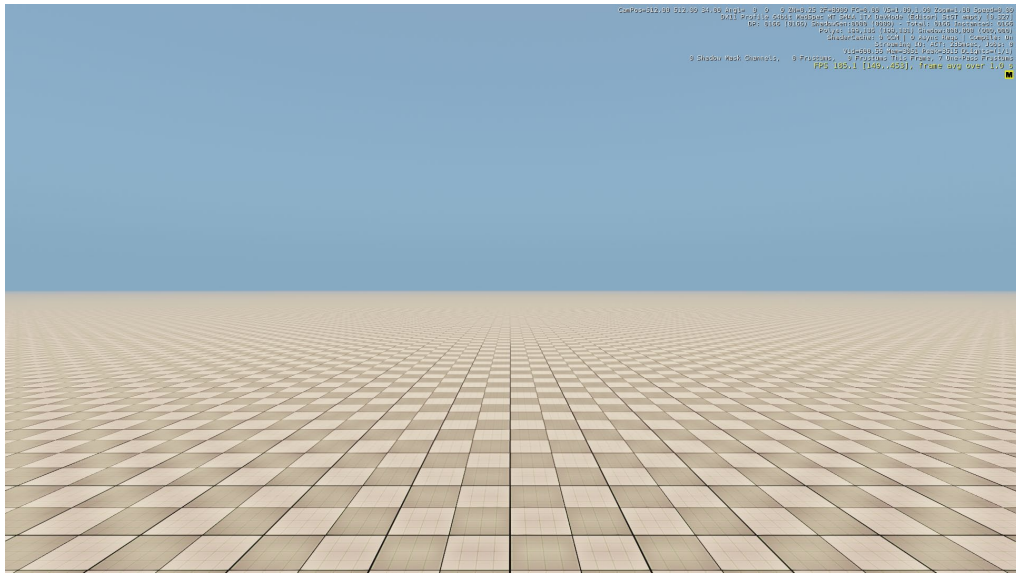
To **look** around without moving, **drag** the **right mouse button**.

To actually **move** the default camera, use the **WASD** keys to move forward, left, backwards, and right respectively. You can also hold down two keys at the same time, for example **W** and **A** to move forward and to the left simultaneously. Use the right mouse button at the same time to control **where** you're moving to. Add **shift** to move faster, or use the scroll wheel to change the camera speed while moving. To set the camera speed without having to move, click on the **Camera** menu at the top left of the Perspective viewport.

If you move past the terrain heightmap, you'll see the **ocean** that's hiding underneath it. You can control what is visible in the Editor, including the ocean and terrain itself using the Perspective Viewport's **Display** menu.

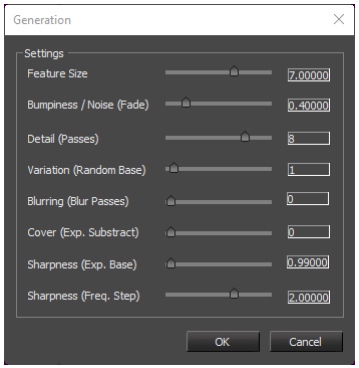
Drag the **middle mouse button** to move straight **up** and **down**. (You can toggle **Camera-Terrain Collisions** - permission to go through the terrain - on or off in the Perspective Viewport's **Camera** menu.)

Finally, you can jump to any position in your level using **Level** > **Go To Position**. (This can also include camera rotation.) All coordinates are in meters. Since your level is 1024x1024 meters, 512 on X and Y are dead center. Note that unlike some 3D modeling programs, CRYENGINE uses Z as the vertical axis.



Introduction: Terrain and Environment

While many game worlds can be quite small or not bear any resemblance to the world we know, and therefore have no need of terrain, let's learn how to give this flat terrain some hills and valleys. There are three ways to do this:



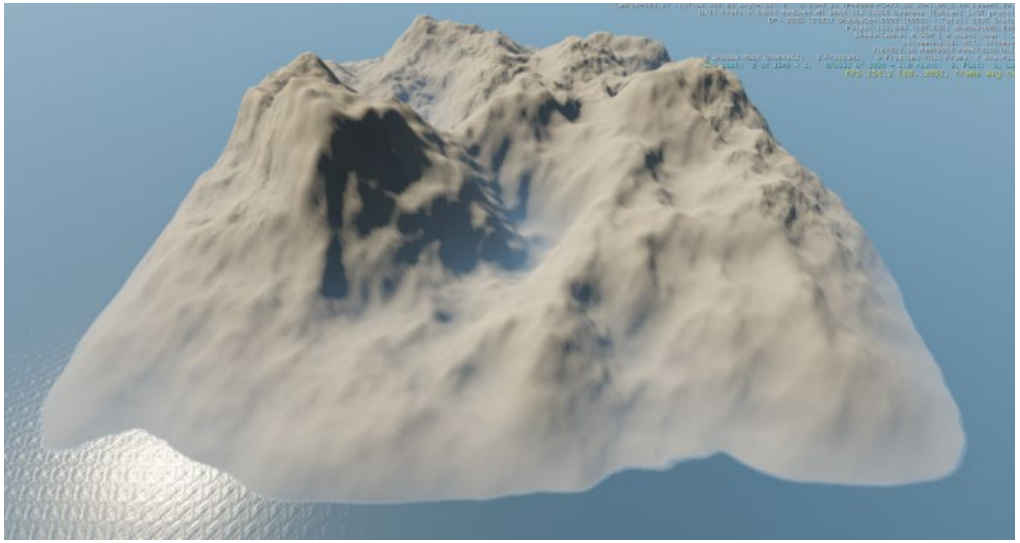
Generating Procedural Terrain

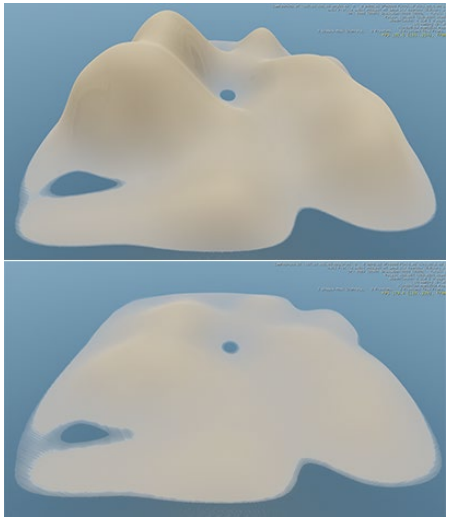
While game developers typically plan their terrain very carefully according to their Game Design Document and how they want to guide players through the level, you can have the engine generate procedural terrain, and either use it as-is or hand-sculpt it as the basis for your level:

From the Terrain Editor, choose **File** > **Generate Terrain**. Read all about the effects these parameters have here. Suffice to say that modifying these values will have a profound effect on the look of your terrain, generating everything from very gentle, featureless hills to near-vertical spikes and everything in between.

For now, accept the default values. You'll end up with the mountainous island shown here. (Move your camera up and back to see the entire terrain.)

This terrain is to steep because the Generate Terrain tool automatically sets the maximum terrain height for a level 1024x1024m wide to **272m**. The procedural terrain generator does this to generate consistent terrain models across most of the range of map sizes, from 128m to 8192m. You can look up the exact heights set by this tool on this documentation page if you're curious.





You can experiment with the settings in the Generate Terrain dialog. Let's try making some gentle hills using the parameters shown here. The result is much more smooth, but the hills are still pretty steep from the perspective of a player walking around.

Let's lower them: in the Terrain Editor, choose **Edit** > **Reduce Range (Heavy)**. Each time you run this, the height of the terrain will get compressed downward, making your hills more gentle. Our example shows the original terrain and then after five applications of this filter.

Adjusting the Lighting

By default, a new level is essentially placed at Earth's equator at high noon, which means that the sun is directly overhead.

In the Environment Editor, experiment with the **North Pole–Equator–South Pole** slider (i.e. latitude) and the **time of day** until you like the look. There is much, much more that the Environment Editor can do, but just changing these two values will have a profound impact on the sunlight in your level. We've used a latitude of 73 and a current time of 17:34.

Another setting that will make a big difference in the realism of your lighting is found in the **Level Settings** tool, which contains additional environmental settings. In the Env State section, there is a setting called **Sun Shadows From Terrain**, which is off by default. It controls whether the terrain casts shadows onto itself using the sun. Turning it on will prevent strange phenomena like your vehicle or vegetation casting shadows while the hill on which they're sitting does not.

Lastly, as you place the sun at a more dramatic angle, the contrast in the lighting may get so high that you can't see into the shadows. You can have the engine add reflected ambient light to your scene, called global illumination: in the **Level Settings** tool

under **Total Illumination V2**, enable **Active**. Reflected light will now lighten your shadows. Keep in mind that calculating global illumination is expensive in terms of computer resources and will typically cost you 20fps of performance.

Adding Clouds

CRYENGINE provides fully volumetric, procedurally generated 3D clouds that re affected by wind and the color of the sunlight. To enable them, you type a Console Variable ("CVar") into the Console tool: `r_VolumetricClouds 1`. Console variables are a way to control the engine directly.

Voilà: clouds appear, whose properties you can explore and control through the **Environment Editor > Volumetric Clouds** settings. For example, if you want the clouds to be affected by the wind, change the Wind influence parameter to a non-zero value.

Sculpting Terrain and Adding a River

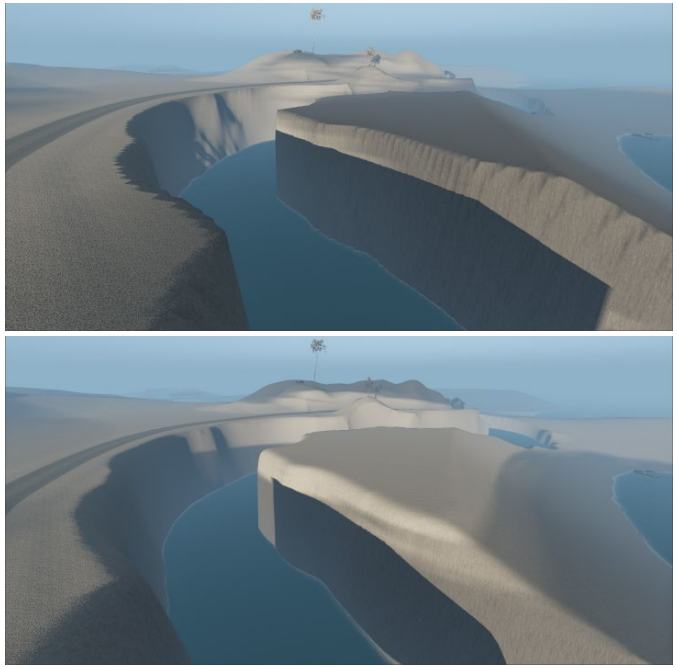
We can also hand-sculpt terrain using the brush tools in the **Sculpt** panel of the **Terrain Editor**. Let's carve a winding canal or two through our landscape. While there is more than one way to accomplish this (including a spline-driven River entity and water bodies), an easy way is simply to lower the terrain so it's below the height of the ocean, which is **16m** by default. (You can set it as you like in the Terrain Editor under **Edit** > **Set Ocean Height**, including negative numbers.)

The Terrain Editor provides two tools that will work: **Flatten** evens out the terrain to a height you choose, creating flat areas. **Raise/Lower** can accomplish the same thing, but it changes the height of each heightmap point **relative to** its current position, whereas Flatten moves all point equally to an **absolute** Z position.

Let's set our flatten tool to an **Outside Radius** of **25m** and an **Inside Radius** of **22m** to give its banks a somewhat gentle slope. (Making these equal will result in vertical banks – sheer cliffs.) Setting the **Height** to **12** means the river bottom will be 4m below the ocean's surface. Use the **Flatten** tool to carve some winding rivers as you like through your level. Since we're moving the heightmap below sea level, our canal will be flooded by the ocean automatically. To make wider, more gentle banks, as in a typical mouth of a river, increase the **Outside Radius**.

Smoothing the Terrain

You may have noticed that where you've used the **Flatten** tool, you have some jagged edges that don't necessarily look realistic. The **Smooth** tool evens out differences in the heightmap. If you painted with it enough, you'd flatten your terrain. Use it selectively, and it's a useful tool to get rid of unwanted artifacts and spikes like those shown here, before and after smoothing. **Radius** is the size of your brush in meters, and **Hardness** is the strength of the brush stroke from zero to 100% (1). Try using it to make your steep edges more realistic, as we've done here.



Terrain before and after smoothing

Importing External Heightmaps

In addition to generating procedural terrain and sculpting it by hand, you can also import an external heightmap. You can read additional tutorials on how to import heightmaps created by third-party procedural terrain generation tools including World Machine. We are also adding tutorials on importing terrain data from **World Creator** and adapting real-world LIDAR or satellite data to an importable heightmap.

Let's import the example heightmap provided with your course assets:

First, let's control how high the highest point on the imported heightmap will be: from the Terrain Editor, choose **Edit > Set Terrain Max Height** and set it to **100m**.

From the Terrain Editor, choose **File** > **Import Heightmap**. Browse to the Getting Started page in the CRYENGINE documentation or open the folder where you downloaded this course, and inside the **sources** folder, choose **practice_heightmap_1K.raw**. You can take this heightmap and sculpt it further, or undo if you prefer to sculpt or import your own terrain. Save your level with **Ctrl-S**.

Seeing the Default GameSDK Setup

It's useful to know what GameSDK actually is. GameSDK contains assets and func-

tionality from Crysis 2, a first person shooter game. This includes thousands of objects, materials, textures, particle effects, sounds, and functionality to help you learn the engine quickly using existing assets. Let's see the player character that GameSDK provides:

Make sure your camera is close to the terrain somewhere (otherwise you'll fall to the ground and die when the game begins!) and press **Ctrl-G** now to enter game play mode. You'll immediately see your hands with a gun in front of you. You can move around with the standard **WASD** keys, shoot the gun (**left mouse click** on a PC), jump (**spacebar**), and toggle between 1st and 3rd person views by pressing **F1**, among other things. It's crucial to test throughout the level design process to get a sense of scale relative to your player.

GameSDK gives you all of this functionality (and much more by default) to help you quickly create prototypes and learn the engine without facing the task of creating everything from scratch. However, if you plan to create all of your own assets and functionality, there is no reason for you to use the GameSDK project.

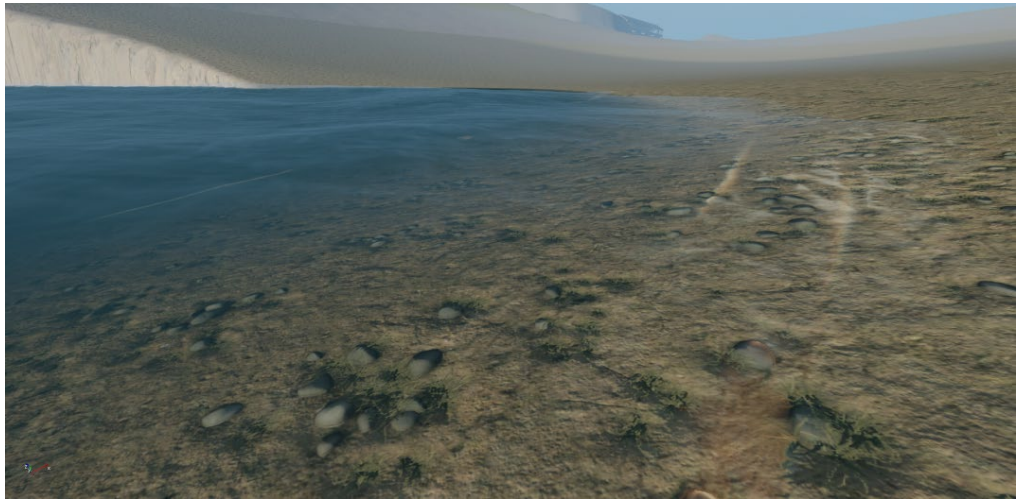
Texturing the Terrain

Our terrain is starting to get interesting, but it needs realistic surfaces. To accomplish that, we use the **Paint** tools in the **Terrain Editor** to add terrain layers. Terrain layers can be filtered by altitude and slope. Let's begin by assigning a mud material to the Default layer to appear the lowest altitudes, including underwater:

Double click the **Default** layer and rename it "mud." Click on the **browse** button next to **Material** and choose **materials/terrain/mud_wet_grassy.mtl**. Also, keep in mind that the texture is what you see at a distance - the general color of the terrain - and the material is the detail you see up close. Note that terrain paint layers must use terrain materials, which use a special terrain shader. Set the **Min Height** to **0**, **Max Height** to **20**, **Min Angle** to **0**, and **Max Angle** to **30**. Click the swatch next to **Filter Color** and choose a nice dark brown muddy color. We've used HTML #6b6050. The filter color tints the material.

Make sure **Mask by Altitude and Slope** and **Paint Layer ID** are enabled. You can use the **Paint** brush to apply a terrain material selectively. However, if you just want to apply it to the entire level, right-click on the mud layer and choose **Flood Layer**. After a few seconds, you should have the mud and rocks texture everywhere up to 25m on the heightmap on any surface up to a 30º slope.

Right-click in the Paint window and create another layer called "grass." Make the filter color a realistic green with a value of about 128. Set the other parameters as



The mud layer at the ocean's edge

follows:

Min Height	30
Max Height	70
Min Angle	0
Max Angle	40
Material:	materials/terrain/grass_4.mtl
Texture:	textures/terrain/detail/grass_4_diff.dds

Again, **right-click** on the layer and choose **Flood Layer**. Note that a terrain layer Material provides the detail that is seen up close, whereas the Texture provides a rough approximation of the layer seen at a distance.

Let's create a layer of dry sand between the wet mud and grass. Create a new layer called "sand" and set its parameters as follows:

Min Height	20
Max Height	30
Min Angle	0
Max Angle	30
Material:	materials/terrain/sand_no_stones.mtl

Again, **right-click** on the layer and choose **Flood Layer**.

Lastly, let's make our steep terrain rocky, like cliffs: create a new layer called "rock"

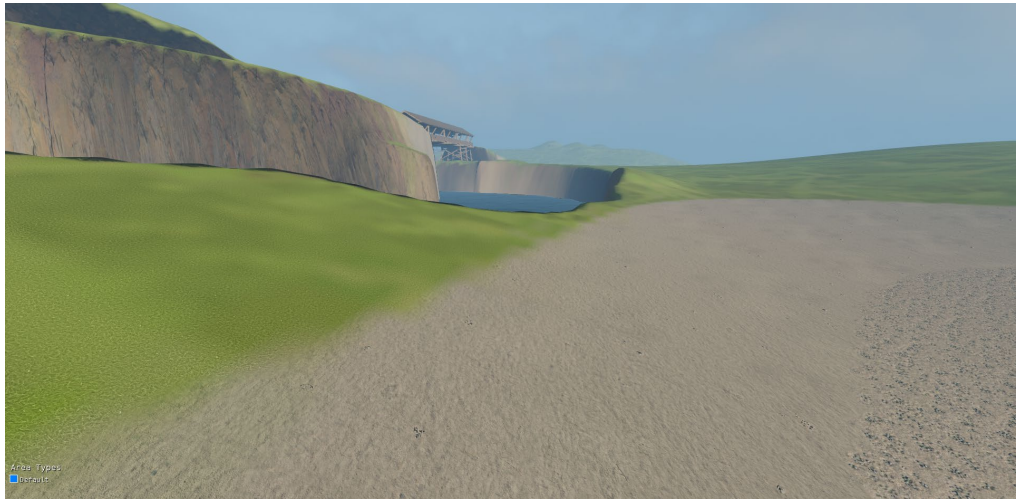
and set its parameters as follows:

Min Height	0
Max Height	1024
Min Angle	40
Max Angle	89.989998
Material:	materials/terrain/canyon_rock_terrain.mtl

Again, right-click on the layer and choose **Flood Layer**. Save your level.

Lastly, you must re-generate the terrain texture after modifying or adding paint layers: from the Terrain Editor's **File** menu, choose **Generate Terrain Texture**. Accept the defaults and click **OK**. Save your level.

If you take a good look around, you'll see that you have the basis of realistic terrain. However, this is just the beginning. Making believable terrain with mixed surfaces calls for additional textures and features on the landscape - vegetation, trees, bushes, rocks, and a fair amount of chaos and variety.



Introduction: Adding Features to the Terrain

To have a truly realistic world, you'll need to add vegetation and inconsistencies in the terrain. Before we add the big items, let's use Decals to hide seams and repetition in the terrain materials.

Using Decals

Decals are small textures including an alpha channel (transparency), designed to be blended on top of other textures. They add realism and details and help hide tiling in textures used on large areas. In the example below, we've used materials/decals/dirt/decal_burned_22 around the campfire to add a burned feeling to sand painted with a terrain layer.

Use **Create Object > Misc > Decal** to add one. In the toolbar, you'll see four "gizmos," any of which can be used to select entities and modify their position, orientation, and size: **Select (4), Move (1), Rotate (2),** and **Scale (3)**. (Their keyboard shortcuts are in parentheses.)



Use the **Move** tool to select the decal. You'll see three arrows, one for dragging the object along each of the three axes, as well as three dual-axis planes to drag an object along two of the three axes.



Drag the decal to whatever surface you want to apply it to until you see the red "Replace Me" symbol. Then click on the **General** > **Material** property in Properties and browse to **Materials** > **Decals** and find one you like. Adjust its position relative to the terrain or object surface to adjust **blending opacity**. You can also rotate a decal non-parallel to the surface to which it's applied to make one edge softer than others.

The other aspect of blending decals into other textures is to check the **Decal > Sort Priority** parameter (0 - 255), which controls the order in which decals are rendered (i.e., what's on top). Decals with higher numbers are **above** those with lower numbers. Decals are a very useful way to add variety to terrain, roads, floors; to stamp words onto walls or doors; to indicate damage dynamically, and much more.

Adding Grass

While you can always place individual .CGF models of trees, bushes, etc. on your terrain, the Vegetation Editor makes it far easier to populate your terrain with larger numbers of assets and to manage their properties.

Choose **Tools > Vegetation Editor**. Let's start with some basic grass:

Click on the **Add Group** button (the leftmost button that has the trees with the + sign inside a box). **Double click** the new "Default1" layer and rename it "grass."

Click the checkbox next to your group to make it visible, and click on the name to select it. Click on the **Add Object** (second) button – the trees with the + sign inside the circle. Browse to **objects/natural/ground/grass_tall/bending_grass_normal_a.cgf**.

Make sure your grass object is selected in the Object list, and set its properties as follows:

Size Var	.3	Vary the size randomly by up to 30% (a recommended value)
Random Rotation		Enabled – helps hide the repetition of the same object
Align to Terrain Normals		Enabled - akes vegetation sprout perpendicular to the terrain. Use with caution for realistic results.
Bending	0	Bending makes the vegetation bend from the force of the wind. Expensive for the engine, so use with discretion, usually on taller vegetation like trees or tall grass.
Density	4	The distance between objects
Elevation Min	30	

Elevation Max	70	
Slope Min	0	
Slope Max	70	(here we've matched the slope and altitude masking we set on the grass terrain layer)
Auto Merged		Enabled - best practice is to always use this for grass (and nothing else) to use a tiled approach to make grass rendering perform better.

Set your Brush **radius** and paint as you like on the terrain.

Adding Trees

Next, let's use another vegetation group to add some trees: Click on the **Add Group** button. Name your new group "trees."

Click the checkbox next to your group to make it visible, and click on the name to select it. Click on the **Add Object** button. Browse to **objects/natural/trees/aspen/b_aspen_a.cgf**. If you want to try adding additional versions of the aspen trees from this folder now, you can, but you will want to differentiate the density and potentially other parameters to keep it realistic.

Make sure your trees group is selected in the Object list, and set its properties as follows:

Size Var	.3	Creates a 30% random variation in size (recommended)
Random Rotation		Enabled - helps hide repetition of same model
Align to Terrain Normals		Disabled. In some case, as with grasses, you may want thi enabled for vegation that should grow perpendicular to the ground plane.
Bending	0	1 (so your trees move in the wind)
Density	35	Since these are large trees, we'll put more space between them (remember units are in meters)
Elevation Min	30	Here our trees will start 14m above the default waterline of the ocean, leaving us some space for terrain typically found closer to the water, like a sandy beach.
Elevation Max	70	At some point, we might want to limit the maximum altitude of certain trees, and give way to those that grow on higher terrain, like non-deciduous evergreens.
Slope Min	0	
Slope Max	30	It wouldn't make sense for the big trees to grow on extremely steep hills.
Auto Merged		Disabled. Only relevant for assets set up to make use of this feature. Read this tutorial for detailed information.

Set your Brush radius and paint as you like on the terrain. If you overdo it, use the **Erase** brush.

Pro Tip:

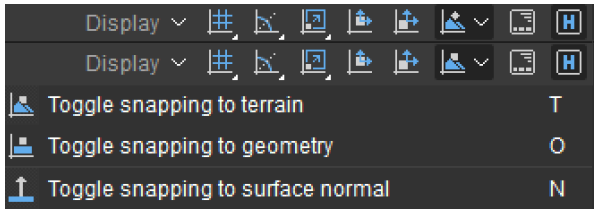
You can **ctrl-click** with the Vegetation Editor to place a single entity.

Adding a Vehicle

GameSDK also provides several vehicles and the animations and functionality to allow the player character to drive them. Let's add one of them:

Using the **Create Object** tool, select **Legacy Entities > Vehicles > HMMWV**. You can either drag the entity onto the terrain or **double-click** it in **Create Object** and then click on the terrain to place it.

Since your terrain is not flat, let's make it easy for your vehicle to stay parallel to the terrain surface, as if its mass kept it naturally on the ground: at the top right of the Perspective Viewport, **click** on the third button from the right. The menu shown in this graphic will appear. Enable both **snapping to terrain** and **snapping to surface normal**. When both are enabled, the icon will look as it does in the top row of this screen shot. Notice the subtle difference in the orientation of the cube floating over the mountains in the icon: parallel to the terrain (i.e., snapping to terrain and surface normal enabled) or not (snapping to surface normal disabled, as shown in the second row in this screen shot). Now you can use the move tool to slide your vehicle along the terrain, and it will remain parallel to its surface. You can also select an entity and **CTRL-SHIFT-click** anywhere on the terrain to snap an object to that position without having to use terrain snapping.



Save your level and then use **Ctrl-G** to enter your game. Walk up to the driver's door of your vehicle and press the **F** key to get into the driver's seat. You can use **F1** to switch between 1st and 3rd person perspective. Use the mouse to change the camera

position when in 3rd person view, and the **WASD** keys to drive. **Spacebar** will brake. Try driving around your level to get a feeling for its scale and bumpiness relative to the vehicle. Think about where you might want to place a road through the level.

You may also want to try the CRYENGINE speedboat, which you also add using **Create Object > Legacy Entities > Vehicles**. Place it in the water so the waterline is just below the red line around the edge of the boat. To drive it, wade into the water, swimming (you'll find that your player character will automatically switch to swimming mode when his feet can no longer touch the bottom) and once you've swum up next to the driver's side of the boat, press the **F** key to get in, just as you do with other vehicles. **WASD** will drive and steer, but beware - this boat is fast!



If you try to drive the boat outside the edges of your terrain map size, you'll quickly learn a memorable, fundamental fact about the engine when your boat flies into the air: **physics are only fully enabled within the boundaries of your terrain heightmap** - thus we refer to it as the **playable terrain**. You can add entities to decorate your world beyond your terrain boundaries, like mountains or buildings - as long as those entities don't need to move or respond to physical forces.

You can use these and other vehicles, their associated animations and scripts in your own games by replacing the models and modifying the behavior scripts as desired.

Adding a Road

Let's give your vehicle a road to drive on. While using the Road tool is pretty straightforward, building roads that are actually drivable takes a fair amount of planning and practice. A few general tips are to avoid sharp turns and steep inclines, to make it wider than you think you need to, and to break long road splines into shorter segments. Take a good look at your terrain and decide where a realistic road could be placed before starting.

As you're about to see, the road tool can be used to alter the heightmap, carving canyons or raising the terrain up to meet it. Before you start drawing your road spline, find a place where a bridge can be placed to span the river you made. Also, it's useful to turn on **Snap to Terrain** to insure that each point you place actually lands on the terrain.

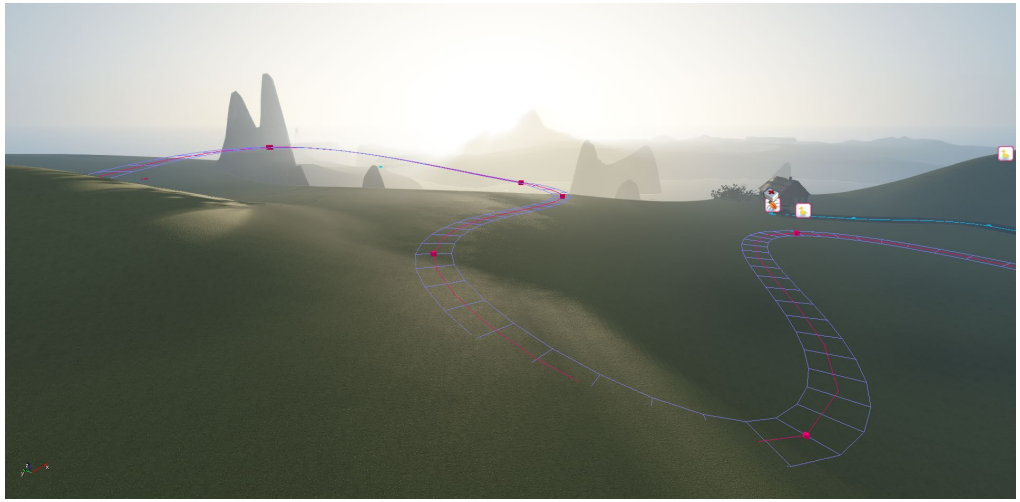
First, we use the road tool to draw a spline curve: using the **Create Object** tool, choose **Misc > Road**. Each time you click on the terrain, a spline point is placed on the terrain. If you click and drag at the same time, you can manually control the curvature at a point. Curvature cannot be adjusted after a point is created, although you can delete and add new points.

Don't make the road too long, but break it into separate segments if necessary. When you get to the edge of your river, **double click** to make the final point.

Before you do anything else, go to the **Properties** window and set the desired **Width** of your road in meters. Try 8m. You'll see the spline adjust its width.

Move your camera around and take a careful look at the points along your road. For ase of editing a road spline point by point, it's often useful to turn off **Snap to Terrain** and use the **Move** gizmo to move individual points.

Sometimes the fastest thing to do if your terrain is fairly flat is to use the **Scale** gizmo



to reduce the variation in the height of your road spline along the Z axis. You will also almost certainly want to click on the **Edit** button in the Properties window, **select** individual points and move them into more precise positions, **delete** those you don't want or need to change, and **Ctrl-click** on the spline to add new ones. Don't forget to click on the **Edit** button again to end spline editing once you're done.

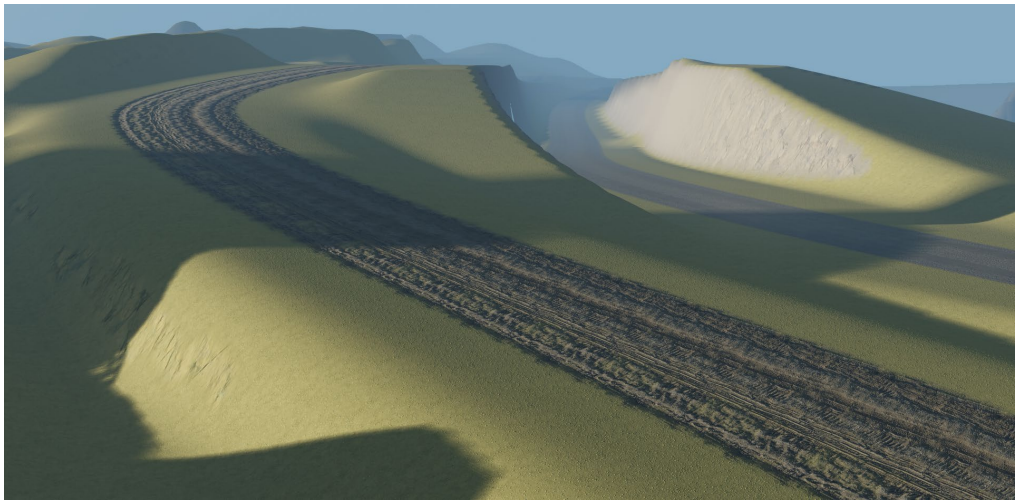
If you want to carve a road into the side of a mountain or steep hill, make sure the outer edge of your road spline is aligned with the surface of the hill, letting the inside edge disappear into the terrain. That way when you align the heightmap to the road surface, you'll be digging into the terrain, which usually looks more realistic then building up steep terrain beneath the road. (For that, you may want to use a bridge of some kind instead of a road spline.) You can also vary the **Angle** of individual points to create **banked turns**.

Pro Tip:

When editing individual points on a road or river spline, make sure your snap to terrain is **off.**

Now let's add a **material** to the road: with the road spline selected, go to the **General** section in the **Properties** panel and click on the **browse** button next to Material. Browse to the **GameSDK > materials > roads** folder and pick something that interests you.

Now comes the dramatic part: with the road spline selected, click on the **Align Height Map** button in the Properties window. The terrain will be modified to follow the road spline, and the terrain painted with the road material. After this step, check your road carefully. You may want to undo the heightmap alignment, adjust individual points, and re-align. You can also vary the Angle property of individual points to create banked turns that are easier to drive at high speeds.



If you have vegetation growing on top of the road, you'll also want to click on the **Erase Vegetation** button, adjusting the **Border** and **Erase Veg Width** parameters as desired. Making Erase Veg Width Var a non-zero value will add some more naturalistic randomization to the edge of the vegetation (unless you're creating a perfectly maintained lawn).

Note that the road and river tools essentially use the same kind of spline tool.

Adding Architectural Elements

Static objects that won't move and with which the player cannot interact are called "brushes." Since they don't require physics calculations, they are very efficient for non-interactive entities. Keep in mind that they also can't be controlled or animated through Flow Graph. Think of a brush as painting 3D objects. Let's use them to add a house and some furniture:

Let's add a house: make sure **Snap to Terrain** is enabled, but not **Snap to Terrain Normals** (assuming you want your floors to be level rather than leaning). Find or sculpt a nice flat spot by the water. From the **Create Object** tool, choose **Brush**. Browse to **objects/architecture/buildings/fishing_houses/fishing_house_c.cgf**.

Pro Tip:

A fast way to add brushes is simply to **drag** meshes (.CGF files) directly from the Asset Browser into a level.

If your house isn't exactly where you want it, you can use the gizmos to **move**, **rotate** and **scale** it. You'll notice that each tool represent the X, Y, and Z axes in a different color. The Rotate gizmo also has as a yellow line that lets you rotate freely about all axes simultaneously. The Move tool also provides planes that let you slide objects along two axes simultaneously, or you can drag any arrow to move an entity along a single axis.

You may find that you have vegetation growing through the floor of your house; if so, use the Vegetation Editor's **Erase** brush to get rid of it.

Pro Tip:

When you're browsing for an asset, there is always a **search** box at the top of the browser. Try typing a word, like "chair".

Once you have your house in place, you can have some fun furnishing it by using Brush entities to place furniture and props that you'll find in the **GameSDK\objects** folder. You'll also find stone walls, fences, piers, bridges, and much more to bring your world to life. In this example, we've



added chairs, a table, refrigerator, and even a stuffed trophy for the wall of our cabin.

Also, as you're placing an entity, you can use the **scroll wheel** on the mouse to **rotate** your entity as you're placing it. **Snapping to Angle** (i.e., rotation) will affect this, whether on or off. Also, use **Snap to Terrain (T)** when you want objects to land on the ground, but turn it off and enable **Snap To Geometry (O)** if you want entities to land on existing objects, like when you're placing a prop on a table.



Soft Depth Blending

One of the most useful features for integrating objects into the terrain believably is called Soft depth test. It blends objects into terrain materials to hide hard edges:

From the **Tools** menu, open the **Material Editor**.

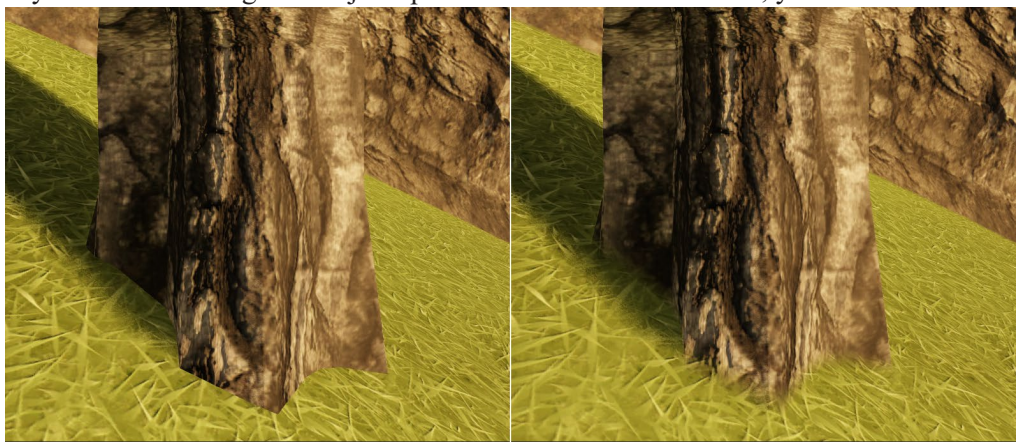
From the Material Editor's File menu, choose **Choose Material From Scene** and click on any terrain material in your level. (Under the eyedropper, you'll see the material name appear.)

The material will be opened for editing in the Material Editor.

Scroll to **Shader Generation Params** and enable **Soft depth test**.

Save your material in the Material Editor menu: **File > Save**.

If you look at the edges of objects placed on this terrain material, you'll see a marked



difference in the way they blend into the terrain. Like all such features, this has a performance cost, so use it only as needed.

Adding an Environment Probe

A simple way to improve the look of your lighting is to add an **environment probe**. Environment probes produce more accurate shadow colors, ambient diffuse values, and specular reflections. Probes also calculate reflected lighting from entities in your

level and factor them into the color of materials on nearby surfaces.

Every level should at least have a global environment probe that covers the entire level. You'll probably find that you'll want to add more localized ones to interiors and specific spaces.

A global environment probe should be placed at the horizontal center of the level with the bottom of the box below the terrain and top at least as high as eye level. The recommended settings for a **global** environment probe are as follows:

Box Size XYZ	Large enough to cover the entire level
AttentuationMaxFalloff	0
IgnoreVisAreas	True
SortPriority	0

Use the **Create Object** tool to select **Misc > Environment Probe**. Place it in the middle of your level, and adjust its properties as detailed above. Environment probes are invisible to the player, as they have no mesh, so keep in mind that you size them using the **Box Size X/Y/Z** parameters, not Transform > Scale.

Note that if you change your level design, you should **recalculate** your environment probes by clicking on **Cubemap > Generate > Cubemaps** in the **Properties** window, as they are not dynamic.



Adding Weather

While plenty of games take place in very constrained, designed environments with no resemblance to the natural world, you can easily add realism and excitement to natural environments using the rain, snow, lightning, and even tornadoes (in addition to the wind built into the Environment Editor) that Game SDK provides.

Try adding the rain effect by using the **Create Object tool > Legacy Entities > Environment > Rain**. Using the **Properties** window > **Lua properties**, you can affect the speedc, intensity, visibility (lighting), and more. You'll notice that it's not just a raindrops overlay; it also places puddles on the ground, splashes rain drops in those puddles in real-time, and makes existing materials look wet using animated decals.

Adding Life

The last thing that we'll do is add some life to our terrain using animated animal assets from GameSDK:

Move to an area where you have beach terrain next to your river.

Using the **Create Object** tool, choose **Legacy Entities > Boids > Frogs**. Click at the water's edge to place your frogs. They will start their froggy behavior immediately, avoiding you if you move near them. You can explore Properties that will affect their behavior including how many frogs per entity and how far they can roam from the entity's center. They will also go in and out of the water.

You'll also find additional "boids" including seagulls, turtles, chickens, etc. with specific associated behaviors which you can use to quickly add life to your levels. You can even make them follow the player.

Summary

This completes our introduction to CRYENGINE fundamentals. You will be using the techniques covered in this brief tutorial in all of your future game designs.

Going Further

If you're ready to learn how to build an actual working game, we suggest that you continue with our extensive free "flappy boid" course, which you can download from our Marketplace. We suggest that you watch the video tutorials of the course and refer to the course workbook for additional information, and to the high resolution screen shots included with the course that clarify the game mechanics and design.